

Problem 16.2

An 8-year annuity due has a future value of \$1,000. Find the periodic payment of this annuity if the effective annual interest rate is 5%.

Solution

Let X be the periodic payment. Since this is an annuity due, payments are made at the beginning of each year. The future value is

$$1000 = X\ddot{s}_{\overline{8}|i}.$$

For an annuity due,

$$\ddot{s}_{\overline{n}|i} = (1+i)s_{\overline{n}|i}.$$

Using $i = 0.05$,

$$\ddot{s}_{\overline{8}|} = (1.05) \left(\frac{(1.05)^8 - 1}{0.05} \right).$$

Thus,

$$1000 = X(1.05) \left(\frac{(1.05)^8 - 1}{0.05} \right).$$

Solving for X ,

$$\begin{aligned} X &= \frac{1000}{(1.05) \left(\frac{(1.05)^8 - 1}{0.05} \right)} \\ &\approx 99.7351. \end{aligned}$$

Hence, the periodic payment is

$$\boxed{\$99.74}.$$